Practical NO.01(A)
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Roll No. -
Title - To use PCA Algorithm for dimensionality reduction. You have a dataset that includes measurements for different variables on wine (alcohol, ash, magnesium, and so on). Apply PCA algorithm & transform this data so that most variations in the measurements of the variables are captured by a small number of principal components so that it is easier to distinguish between red and white wine by inspecting these principal components. Dataset Link: https://media.geeksforgeeks.org/wp-content/uploads/Wine.csv

In [3]:
```
import pandas as pd
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
import matplotlib.pyplot as plt
print("All standard libraries imported successfully.")
```

```
All standard libraries imported successfully.
```

In [4]:
```
# The URL must be passed as a variable or wrapped inside quotes ""
url = "https://media.geeksforgeeks.org/wp-content/uploads/Wine.csv"
print("Downloading the official manual dataset from the internet...")
df = pd.read_csv(url)
print(f"Dataset successfully loaded into memory! Shape: {df.shape}")
df.head(5)
```

```
Downloading the official manual dataset from the internet...
Dataset successfully loaded into memory! Shape: (178, 14)
```

Out[4]:

	Alcohol	Malic_Acid	Ash	Ash_Alcanity	Magnesium	Total_Phenols	Flavanoids	Non
0	14.23	1.71	2.43	15.6	127	2.80	3.06	
1	13.20	1.78	2.14	11.2	100	2.65	2.76	
2	13.16	2.36	2.67	18.6	101	2.80	3.24	
3	14.37	1.95	2.50	16.8	113	3.85	3.49	
4	13.24	2.59	2.87	21.0	118	2.80	2.69	

In [5]:
```
# Prints unique segments and missing entry analysis as requested on Page 10 of t
print("Unique Customer Segments available:")
print(df['Customer_Segment'].unique())
print("\nMissing values check per column:")
print(df.isnull().sum())
```

```
Unique Customer Segments available:
[1 2 3]

Missing values check per column:
Alcohol                 0
Malic_Acid              0
Ash                     0
Ash_Alcanity            0
Magnesium               0
Total_Phenols           0
Flavanoids              0
Nonflavanoid_Phenols    0
Proanthocyanins         0
Color_Intensity         0
Hue                     0
OD280                   0
Proline                 0
Customer_Segment        0
dtype: int64
```

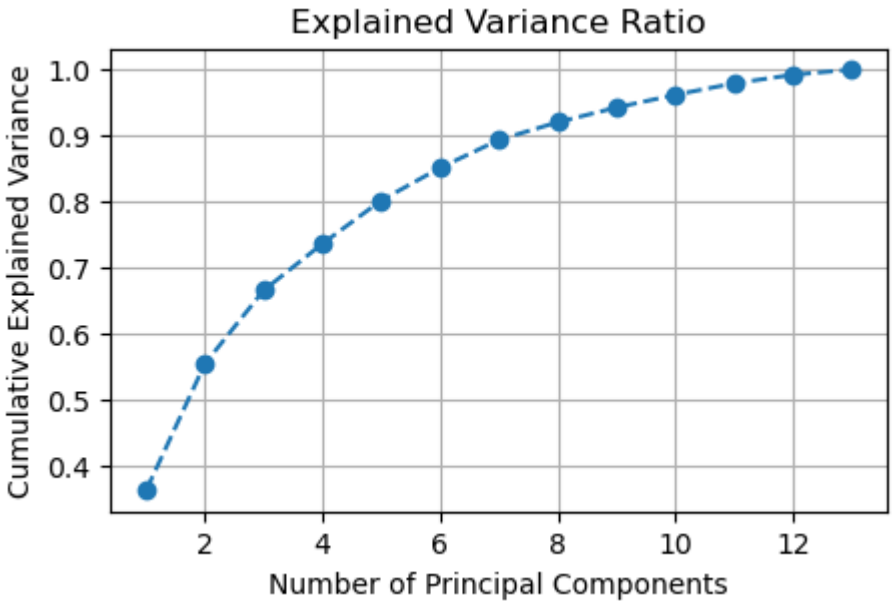
In [6]:
```
# Splits attributes and scales each parameter according to the manual formula
X = df.drop('Customer_Segment', axis=1)
y = df['Customer_Segment']
for col in X.columns:
    sc = StandardScaler()
    X[col] = sc.fit_transform(X[[col]])
print("Features scaled and standard scores computed successfully.")
X.head(5)
```

```
Features scaled and standard scores computed successfully.
```

Out[6]:

	Alcohol	Malic_Acid	Ash	Ash_Alcanity	Magnesium	Total_Phenols	Flavanoids
0	1.518613	-0.562250	0.232053	-1.169593	1.913905	0.808997	1.034819
1	0.246290	-0.499413	-0.827996	-2.490847	0.018145	0.568648	0.733629
2	0.196879	0.021231	1.109334	-0.268738	0.088358	0.808997	1.215533
3	1.691550	-0.346811	0.487926	-0.809251	0.930918	2.491446	1.466525
4	0.295700	0.227694	1.840403	0.451946	1.281985	0.808997	0.663351

In [7]:
```
# Creates the mathematical step to figure out variance profile across components
pca = PCA()
X_pca = pca.fit_transform(X)
explained_variance_ratio = pca.explained_variance_ratio_
plt.figure(figsize=(5,3))
plt.plot(range(1, len(explained_variance_ratio) + 1), explained_variance_ratio.c
plt.xlabel('Number of Principal Components')
plt.ylabel('Cumulative Explained Variance')
plt.title('Explained Variance Ratio')
plt.grid(True)
plt.show()
```



In [8]:
```
# Final evaluation component mapping and spatial sorting plot
n_components = 13
pca = PCA(n_components=n_components)
X_pca = pca.fit_transform(X)
red_indices = y[y == 1].index
white_indices = y[y == 2].index
plt.figure(figsize=(5,3))
plt.scatter(X_pca[red_indices, 0], X_pca[red_indices, 1], c='red', label='Red Wi
plt.scatter(X_pca[white_indices, 0], X_pca[white_indices, 1], c='blue', label='W
plt.xlabel('Principal Component 1')
plt.ylabel('Principal Component 2')
plt.legend()
plt.title('PCA: Red Wine vs. White Wine')
plt.grid(True, alpha=0.3)
plt.show()
```

